Thyristor-Based Soft Starters for High-Power Motors: Principles, Power Ranges, and Snubber Design

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Abstract

This note gives an engineering overview of thyristor-based soft starters (reduced-voltage solid-state starters) for medium and large induction motors. It covers the motor starting problem, the phase-angle control principle, the main circuit topologies (in-line, inside-delta, medium-voltage series strings), the power levels at which soft starters are typically applied, device selection, thermal duty, and the interaction with the supply during the starting ramp. A dedicated section discusses the role and sizing of snubbers in these applications: $\mathrm{d}v/\mathrm{d}t$ limitation of phase-control thyristors, damping of commutation transients, dynamic voltage sharing in series-connected strings, and coordination with metal-oxide varistors. A worked sizing example for a 690 V, 1 MW starter is included.

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1 Introduction

A three-phase squirrel-cage induction motor connected directly to the line (direct-on-line, DOL, starting) draws a starting current of typically 5 to 8 times its rated current at a poor power factor, and develops a starting torque that is fixed by the motor design rather than by the needs of the driven load. For small motors on stiff supplies this is acceptable. For motors in the hundreds of kilowatts to tens of megawatts range, DOL starting causes:

- severe voltage dips on the supply, which may violate utility connection agreements or trip other loads;
- high electrodynamic and thermal stress in the stator winding and in the rotor cage, reducing motor life when starts are frequent;
- mechanical shock in the driven equipment (couplings, gearboxes, belts) and hydraulic transients (water hammer) in pump systems.

The soft starter is the simplest solid-state answer to this problem. It is a three-phase AC voltage controller built with anti-parallel phase-control thyristors, inserted between the line and the motor. By delaying the firing of the thyristors with respect to the line voltage, the fundamental voltage applied to the motor is reduced in a controlled way, and with it the starting current and torque. Once the motor has reached rated speed the thyristors are fired continuously (or the starter is bypassed by a contactor) and the motor runs directly on the line. The soft starter therefore controls only the *starting and stopping transients*; unlike a variable-frequency drive (VFD) it provides no continuous speed control.

Soft starters are attractive at high power precisely because the power circuit is minimal: two thyristors per phase, no DC link, no fast switching, no output filter. The cost, volume, and losses are a small fraction of those of a VFD of equal rating, which is why soft starters remain the standard solution for fixed-speed pumps, fans, compressors, conveyors, crushers, and mills from a few tens of kilowatts up to tens of megawatts.

2 The Starting Problem

For a squirrel-cage motor the locked-rotor current I_{LR} is essentially determined by the total leakage impedance and is nearly independent of the mechanical load:

$$I_{\mathrm{LR}} \approx \frac{V_{\mathrm{ph}}}{\sqrt{(R_s + R'_r)^2 + (X_{\sigma s} + X'_{\sigma r})^2}} \approx (5 \ldots 8)\, I_N. \tag{1}$$

At reduced stator voltage the current scales linearly with voltage, while the torque at any given slip scales with the voltage squared:

$$I(s) \propto V, \qquad T(s) \propto V^2. \tag{2}$$

Equation (2) is the fundamental trade-off of every reduced-voltage starting method: halving the starting current quarters the starting torque. A soft start is therefore only possible if the load

torque during acceleration is well below the motor torque at the reduced voltage. This is the case for centrifugal loads (pumps, fans), where the load torque grows approximately with the speed squared, and it is the reason why soft starters are so common in exactly those applications. High-breakaway loads (loaded conveyors, positive-displacement compressors, mills with material) require a higher current limit or a different starting method.

Figure 1 illustrates the effect: the motor torque–speed curve is scaled by $(V/V_N)^2$, and the intersection with the load curve defines the accelerating torque available during the ramp.

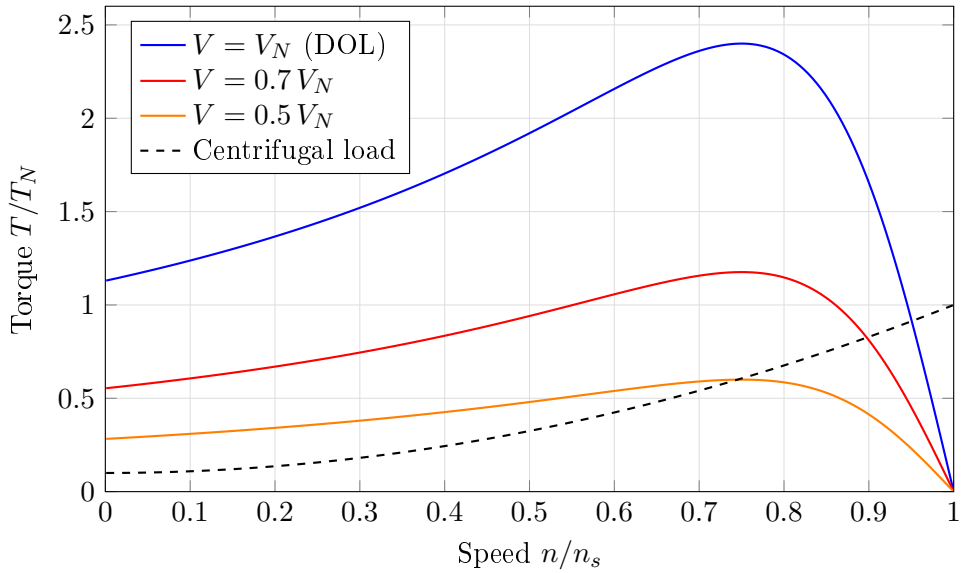


Figure 1: Induction motor torque–speed curves at reduced voltage ($T \propto V^2$) and a centrifugal load curve. At $0.5\,V_N$ the motor still accelerates a centrifugal load, but would stall against a constant-torque load of rated value.

3 Overview of Reduced-Voltage Starting Methods

Table 1 places the soft starter among the classical alternatives. The electromechanical methods (star–delta, autotransformer, reactor) reduce the voltage in one or two fixed steps; the transition between steps produces current and torque transients that can approach DOL values. The soft starter replaces the steps by a continuous, closed-loop controlled ramp, and adds soft stopping, which no electromechanical method provides.

Table 1: Comparison of starting methods for a motor with $I_{\mathrm{LR}} = 6\,I_N$.

Method	Line current	Torque	Transients	Relative cost
Direct on line	$6\,I_N$	T_{LR}	none (single shock)	lowest
Star–delta	$2\,I_N$	$T_{\mathrm{LR}}/3$	severe at changeover	low
Autotransformer	$(2\ldots4)\,I_N$	$k^2\,T_{\mathrm{LR}}$	moderate at steps	medium
Series reactor	$(3\ldots4.5)\,I_N$	$k^2\,T_{\mathrm{LR}}$	moderate	medium
Soft starter	$(2\ldots4.5)\,I_N$, adjustable	$\propto V^2$, ramped	none	medium
VFD	$\leq I_N$	up to T_N at all speeds	none	highest

4 The Thyristor Soft Starter

4.1 Phase-angle control principle

The power circuit is a three-phase AC voltage controller: one pair of anti-parallel phase-control thyristors per phase (Fig. 2). Each thyristor is fired with a delay angle α measured from the zero crossing of its anodic voltage and turns off naturally at the following current zero (line commutation). Increasing α shortens the conduction interval and reduces the fundamental component of the motor voltage. Because the motor is an inductive load with a back-EMF, the relation between α and the fundamental voltage is nonlinear and load-dependent; practical starters therefore close a control loop around measured current rather than commanding a voltage directly.

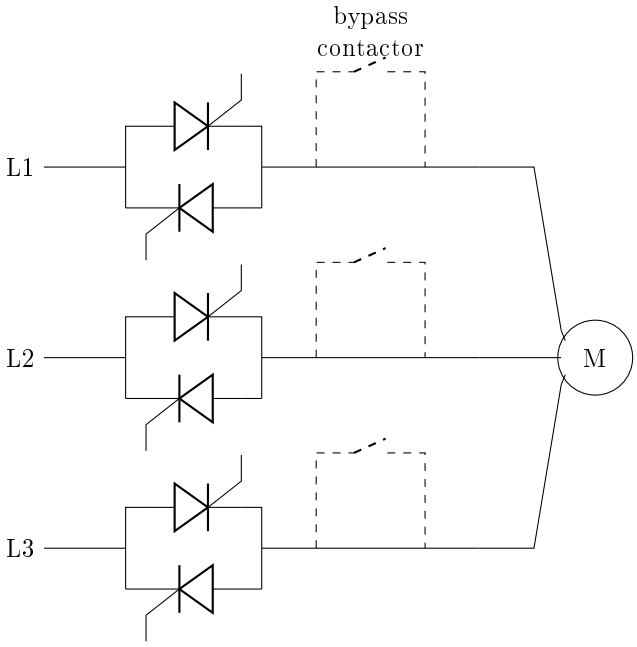


Figure 2: In-line three-phase soft starter: one anti-parallel pair of phase-control thyristors per phase, with optional bypass contactor closed after run-up.

Two properties of this circuit shape the whole design:

1. **Line commutation.** The thyristors never interrupt current; they stop conducting at natural current zeros. The $\mathrm{d}i/\mathrm{d}t$ at turn-on and the reverse recovery are mild compared to inverter service, which is why standard phase-control (converter-grade) thyristors are used, not fast thyristors.

2. **Discontinuous, asymmetric conduction.** During the ramp the phase currents are non-sinusoidal and may be discontinuous. Gating must use pulse trains (or long back-porch pulses) so that a thyristor is re-fired if its current attempts to restart within the intended conduction interval.

4.2 Topology variants

In-line (six thyristors). The standard arrangement of Fig. 2. Each pair carries the full line current and blocks line-to-line voltage levels. This is the topology used at high power and the only one used at medium voltage.

Inside-delta (six thyristors, six motor leads). If all six winding terminals are available, the pairs can be placed in series with each delta winding. Each pair then carries only the winding

current, $I_{\text{line}}/\sqrt{3} \approx 0.58\, I_{\text{line}}$, allowing a smaller starter frame for a given motor. The wiring is more complex and a wiring error (e.g., swapped leads) can produce large uncontrolled currents, so this variant is mostly found in LV retrofit applications.

Two-phase controlled (four thyristors). One phase is connected solidly and only two phases are controlled. This saves two devices but produces asymmetric currents and a torque pulsation during the ramp. It is restricted to small LV starters and is not used in the high-power range considered here.

Bypass. Almost all modern starters close a bypass contactor once the motor has reached full voltage. This removes the thyristor conduction losses (roughly $3 \times V_{T0} I_{\text{avg}}$ plus resistive terms, i.e., kilowatts at megawatt motor ratings), shrinks the cooling system, and lets the thermal design be based on the *starting duty* only. The thyristors must still be rated to carry the start current for the ramp duration and to withstand the prospective short-circuit current until protection clears.

4.3 Control functions

Typical control features of an industrial soft starter:

- **Voltage ramp:** open-loop ramp of the firing angle from an initial voltage (30–60 % V_N) to full voltage in an adjustable time (typically 2–60 s).
- **Current limit:** closed-loop regulation of the RMS line current to a set value, typically $(2 \ldots 4.5)\, I_N$, during acceleration. This is the mode normally used at high power, because it makes the supply loading predictable.
- **Kick start:** a short (0.1–1 s) pulse of higher voltage to break away high-stiction loads before the ramp.
- **Soft stop:** a controlled voltage ramp-down (10–60 s), essential in pumping systems to avoid check-valve slam and water hammer.
- **Protections:** electronic overload (thermal model of the motor), locked rotor, phase loss and phase reversal, thyristor failure (shorted SCR detection), starter over-temperature.

The implementation of these functions — synchronization, firing logic, and the control loop — is discussed in Section 5.

5 Control and Firing Architecture

5.1 Firing infrastructure: the six-pulse bridge heritage

The gating infrastructure of a soft starter is essentially that of a line-commutated six-pulse (B6) thyristor bridge, and early industrial soft starters were in fact built around rectifier firing boards. The common elements are:

- **Line synchronization:** a PLL (or filtered zero-crossing detector) locked to the supply provides the phase reference; it must reject the commutation notches and harmonic distortion that the starter itself injects during the ramp.
- **Six delay channels:** one firing-angle counter per thyristor, loaded from the common angle command and started at the reference instant of the corresponding device. The firing order is a ring counter advancing in 60° steps, exactly as in a B6 bridge (for phase

sequence L1–L2–L3: $T1^+$, $T3^-$, $T2^+$, $T1^-$, $T3^+$, $T2^-$, where $+/-$ denote the two devices of each anti-parallel pair).

- **Double pulsing:** with the standard three-wire connection (no neutral), a phase current can only close through another phase. A thyristor fired alone therefore cannot conduct: at every firing instant the incoming device *and* the appropriate device of a second phase must be gated together — the exact analogue of the second (repeat) pulse in a B6 bridge. Equivalently, each gate command is issued twice, 60° apart.
- **Pulse trains with back-porch:** because conduction is discontinuous during the ramp, a device whose current attempts to restart within its intended conduction interval must be re-fired. Gating therefore uses high-frequency pulse trains (or long pulses with a sustained back-porch current above I_{GT}) covering the whole interval, transmitted through pulse transformers at LV and fiber-optic links to local gate units at MV.

5.2 Differences from the rectifier

Three points prevent a literal reuse of the rectifier control law:

Angle reference. In a B6 bridge, α is measured from the natural commutation points (the crossings of the line-to-line voltages). In the AC controller each thyristor is referred to the zero crossing of *its own anodic voltage*, and this voltage is not simply the line voltage: after a current zero the blocking pair sees the difference between the line and the motor terminal voltage, which contains the residual back-EMF of the still-rotating machine. Robust implementations therefore do not time the delay from the ideal line zero crossing, but synchronize on the measured device current zeros and control the *hold-off angle* γ — the non-conduction interval between the current zero of one device and the firing of the opposite device. Controlling γ instead of α has two benefits: the plant seen by the current loop becomes nearly linear in γ, and the two half-waves are symmetrized by construction, which suppresses DC components in the motor current (a DC component produces braking torque and asymmetric device loading) [5].

Actuator gain. The rectifier has the clean, load-independent characteristic $V_{\mathrm{dc}} = 1.35\, V_{\mathrm{LL}} \cos\alpha$ in continuous conduction. The AC controller has no such closed form: the relation between α and the fundamental motor voltage is nonlinear and depends on the load impedance angle φ, which itself changes along the run-up (from the nearly purely inductive locked-rotor condition, $\varphi \approx 75°$–$85°$, towards the rated power factor). For $\alpha \le \varphi$ the current is continuous and the controller has no authority at all; the useful control range is approximately $\varphi < \alpha \lesssim 150°$. An open-loop α ramp is therefore only a coarse mode; the standard high-power solution closes the loop on measured current.

No commutation overlap. There is no transfer of current between devices, hence no overlap angle u and no inversion limit; each device simply extinguishes at its own current zero. On the other hand, conduction mode changes appear that the bridge does not have: depending on α and φ the circuit moves between three-device and two-device conduction, with the well-known dead band and gain change between modes.

5.3 Control structure

Figure 3 shows the resulting structure. An outer ramp generator produces the current reference (constant current limit, or a profile); a PI regulator acting on the RMS (or rectified-mean, updated every half cycle) line current commands the hold-off angle; the pulse distributor turns the angle command into the six double-pulse gate trains, synchronized by the PLL and by the device current-zero detectors. The half-cycle current measurement makes the loop inherently

sampled at $2f_{\text{line}}$ per phase; the attainable current-loop bandwidth is a few tens of hertz, which is entirely adequate for a process whose time constant is the mechanical run-up (seconds).

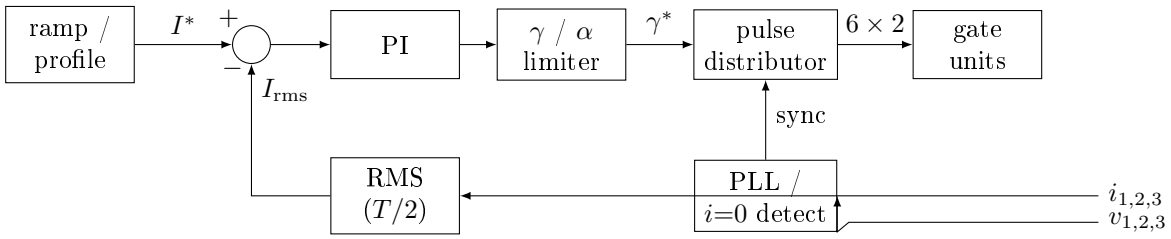


Figure 3: Control and firing structure of a current-limited soft starter. The PI regulator commands the hold-off angle γ; the pulse distributor generates the six gate signals with double pulsing (60°-shifted repeat pulse), synchronized to the line PLL and to the device current-zero detectors.

Additional logic layers found in industrial products: firing symmetry supervision (equal positive/negative half-cycle volt-seconds, anti-DC), kick-start sequencing, transition to full conduction and bypass closure (the thyristors are kept gated while the bypass closes, so the contactor never makes the current), and soft-stop ramp-down with the same current loop. Shorted-thyristor detection is obtained cheaply by monitoring the voltage across each pair: a device that shows no blocking interval within a half cycle is declared failed.

6 Power Levels and Device Selection

6.1 Low-voltage range (200 to 690 V)

LV soft starters are catalogue products from roughly 5 kW up to about 1 to 2 MW (rated currents up to 1600 to 2100 A at 400 V–690 V). The upper bound is set by the practicality of LV cabling and switchgear rather than by the semiconductors. Devices are phase-control thyristors in disc (press-pack) or module packages with blocking voltages of:

- 1200 to 1600 V class for 400 V networks,
- 1800 to 2200 V class for 690 V networks,

i.e., a repetitive blocking voltage V_{DRM} of at least 2–2.5 times the peak line-to-line voltage, so that normal switching surges are absorbed without relying on the varistors.

6.2 Medium-voltage range (2.3 to 15 kV)

MV soft starters cover motors from a few hundred kilowatts to several tens of megawatts at standard levels of 2.3, 3.3, 4.16, 6.0/6.6, 10/11, and 13.8 kV. Since single phase-control thyristors are available up to about 6.5 to 8.5 kV blocking, each "switch" becomes a *series string* of N anti-parallel pairs per phase, usually with $N + 1$ redundancy so that one shorted device does not force a shutdown. For example, a 6.6 kV starter (peak line-to-line 9.3 kV) built with 6.5 kV-class devices and a design utilization of 2.5 kV per device uses 4–5 devices per direction per phase. Series operation is what makes voltage-sharing networks — and therefore snubbers — a first-order design item at MV, as discussed in Section 8.

Typical MV applications: boiler feed and cooling water pumps, induced/forced draft fans, refrigeration and gas compressors, SAG/ball mill auxiliaries, crushers, and large marine or hydro auxiliaries. In many of these, the soft starter is shared between several motors through a switching scheme (one starter, n motors), which is economically attractive because the starter works only a few tens of seconds per start.

6.3 Thermal duty and utilization categories

Because the semiconductors work only during the ramp (with bypass), the rating of a soft starter is inherently a *short-time, repeated-duty* rating. IEC 60947-4-2, the product standard for AC semiconductor motor controllers and starters, encodes this in the AC-53 utilization categories; for example

$$\text{AC-53a: } 3.5\text{–}30 : 50\text{–}10$$

means the starter carries $3.5\,I_e$ for 30 s per start, with 50 % on-load duty and 10 starts per hour. The same hardware carries a higher current rating with a bypass (AC-53b) than without, because the junction temperature can start each ramp from a lower value. At high power the practical limit is almost always the number of starts per hour allowed by the *motor* thermal model, not by the starter.

The relevant device stresses are:

- RMS/average current during the ramp (junction temperature excursion, checked against the transient thermal impedance $Z_{\text{th}}(t)$);
- surge current I_{TSM} and I^2t for coordination with fuses or breakers under motor or cable faults;
- reapplied $\mathrm{d}v/\mathrm{d}t$ and $\mathrm{d}i/\mathrm{d}t$, both mild in line-commutated service but still requiring snubbing, as discussed next.

7 Interaction with Supply and Motor

During the ramp the phase-controlled currents contain low-order odd harmonics (5th, 7th, 11th, ...) and, at large firing angles with discontinuous conduction, significant higher-order content; the starter also notches the supply voltage at the commutation instants. Because this lasts only for the ramp duration (seconds), harmonic standards treat it as a short-duration emission and it is rarely a compliance problem; it can, however, disturb sensitive loads on weak feeders, and capacitor banks on the same bus must not be switched during the ramp (resonance excitation).

On the motor side, reduced-voltage non-sinusoidal supply produces additional rotor losses and, at large α, a torque ripple at $6\times$ line frequency plus possible subharmonic pulsations under asymmetric firing. Modern starters symmetrize the firing between the two half-waves precisely to avoid DC components in the motor current, which would otherwise cause braking torque and transformer-like saturation effects. Extended operation at partial voltage (e.g., misused "energy-saver" modes) increases motor heating and is not recommended; the soft starter should be seen strictly as a starting/stopping device.

8 Snubbers in Soft-Starter Applications

8.1 Why snubbers are needed

Although the soft starter is a line-commutated converter with mild switching stresses, three mechanisms make snubbers mandatory:

1. **Static and reapplied** $\mathrm{d}v/\mathrm{d}t$**.** A phase-control thyristor can be triggered spuriously by displacement current through its center junction capacitance if the anode voltage rises too fast. Datasheet critical values are typically 500 to 1000 V/µs for a blocking device and much lower during the turn-off interval t_q, when reapplied $\mathrm{d}v/\mathrm{d}t$ can re-fire a device that has just commutated. In the soft starter, every current zero of an outgoing thyristor is followed by a step of instantaneous line voltage across the pair; through the supply stray

inductance this step arrives as a fast, ringing edge unless it is snubbed. A spurious turn-on in an anti-parallel pair short-circuits nothing by itself, but it destroys the voltage control, can create DC components in the motor current, and in a series string it overvoltages the remaining devices.

2. **Reverse-recovery (commutation) overvoltage.** At each current zero the outgoing thyristor sweeps out its recovered charge; the recovery current snapping off in the loop inductance produces a voltage overshoot across the device. In line-commutated service the $\mathrm{d}i/\mathrm{d}t$ at the current zero is low ($\omega\hat{I}$, a few A/µs), so I_{RM} is modest, but with the large disc devices used at high power the recovered charge is still tens of millicoulombs and the overshoot must be damped.

3. **Line-borne transients and voltage sharing.** The starter hangs directly on the distribution bus and sees contactor switching surges, capacitor-bank energization transients, and lightning-borne surges. In MV series strings, the snubber additionally performs *dynamic voltage grading*: it equalizes the voltage across devices that turn off with slightly different recovered charges or turn on with different delay times.

8.2 RC snubber across the pair

Since at any instant at most one device of an anti-parallel pair is blocking, a single RC snubber connected across the pair protects both directions (Fig. 4). A metal-oxide varistor (MOV) in parallel clamps energy-rich surges that the RC cannot absorb.

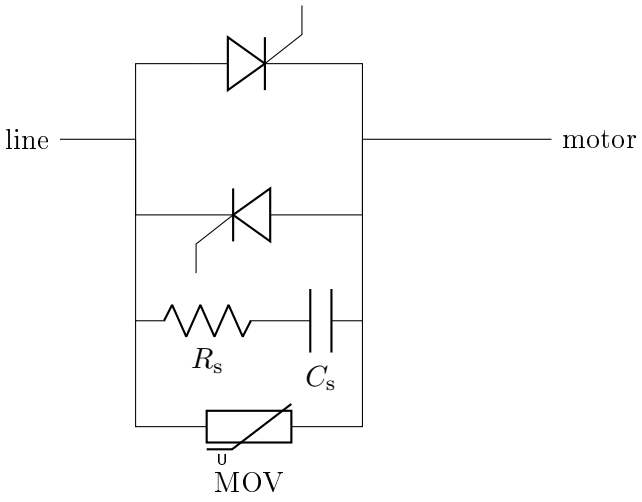


Figure 4: Anti-parallel thyristor pair with RC snubber and metal-oxide varistor. One RC network per pair suffices, since only one device blocks at a time.

Sizing model. The design case is a voltage step V_0 (worst case: the instantaneous line-to-line voltage at a current zero, up to $\hat{V}_{\mathrm{LL}} = \sqrt{2}\,V_{\mathrm{LL}}$) applied through the supply inductance per phase L_σ to the R_{s}–C_{s} branch. With the device blocking, the series $L_\sigma R_{\mathrm{s}} C_{\mathrm{s}}$ circuit gives:

$$\omega_0 = \frac{1}{\sqrt{L_\sigma C_{\mathrm{s}}}}, \qquad Z_0 = \sqrt{\frac{L_\sigma}{C_{\mathrm{s}}}}, \qquad \zeta = \frac{R_{\mathrm{s}}}{2Z_0}, \tag{3}$$

$$\left.\frac{\mathrm{d}v_{\mathrm{T}}}{\mathrm{d}t}\right|_{t=0} = \frac{R_{\mathrm{s}} V_0}{L_\sigma}, \qquad \left.\frac{\mathrm{d}v_{\mathrm{T}}}{\mathrm{d}t}\right|_{\mathrm{peak}} \approx k(\zeta)\,\omega_0 V_0, \tag{4}$$

where $k(\zeta) \approx 1$ for $\zeta \approx 0.5 \ldots 0.7$. The classical optimization (McMurray [4]) trades peak voltage against peak $\mathrm{d}v/\mathrm{d}t$; for soft starters, where the switching frequency is the line frequency and snubber loss is a secondary concern, a simple and robust recipe is:

1. Choose C_s so that $\omega_0 V_0 \le (0.2 \ldots 0.3) \times (\mathrm{d}v/\mathrm{d}t)_\mathrm{crit}$, keeping a large margin because the reapplied-$\mathrm{d}v/\mathrm{d}t$ capability right after commutation is far below the static datasheet value.
2. Choose $R_\mathrm{s} = 2\zeta Z_0$ with $\zeta \approx 0.5 \ldots 0.7$ for near-critical damping of the ring.
3. Check the additional overshoot due to recovery: $\Delta V \approx I_\mathrm{RM} Z_0$, and verify $V_\mathrm{peak} \le V_\mathrm{DRM}$ with margin.
4. Check the snubber discharge current at thyristor turn-on, $\hat{I}_\mathrm{dis} \approx V_0/R_\mathrm{s}$, against the gate area's $\mathrm{d}i/\mathrm{d}t$ capability (this sets a lower bound on R_s).
5. Rate the resistor for the cyclic energy $P_R \approx f_\mathrm{line} C_\mathrm{s} V_0^2$ (charge and discharge once per half cycle per polarity) with a pulse-rated (wirewound or thick-film) part.

8.3 Worked example: 690 V, 1 MW starter

Motor: 1 MW, 690 V, $\eta = 0.96$, $\cos\varphi = 0.85$, hence $I_N \approx 1025$ A; current-limit start at $3.5\, I_N \approx 3.6$ kA RMS for 30 s. Supply: 2 MVA transformer, $u_k = 6\,\%$:

$$L_\sigma = \frac{u_k V_\mathrm{LL}^2}{\omega S} = \frac{0.06 \times 690^2}{314 \times 2 \times 10^6} \approx 45\,\mu\mathrm{H} \text{ per phase.} \tag{5}$$

Worst-case step: $V_0 = \sqrt{2} \times 690 \approx 976$ V. Device: 2200 V-class phase-control disc thyristor, $(\mathrm{d}v/\mathrm{d}t)_\mathrm{crit} = 1000\,\mathrm{V/\mu s}$.

Choosing $C_\mathrm{s} = 0.5\,\mu\mathrm{F}$:

$$\omega_0 = \frac{1}{\sqrt{45\,\mu \times 0.5\,\mu}} \approx 211\,\mathrm{krad/s}, \qquad Z_0 = \sqrt{\frac{45\,\mu}{0.5\,\mu}} \approx 9.5\,\Omega, \tag{6}$$

$$\left(\frac{\mathrm{d}v}{\mathrm{d}t}\right)_\mathrm{peak} \approx \omega_0 V_0 \approx 206\,\mathrm{V/\mu s} \quad (\approx 20\,\% \text{ of the static limit; ample margin for } t_q). \tag{7}$$

Damping: $R_\mathrm{s} = 2 \times 0.5 \times 9.5 \approx 10\,\Omega$ ($\zeta = 0.5$); initial slope $R_\mathrm{s} V_0 / L_\sigma = 217\,\mathrm{V/\mu s}$, consistent. Recovery overshoot with $I_\mathrm{RM} \approx 60$ A (low $\mathrm{d}i/\mathrm{d}t$, large device): $\Delta V \approx 60 \times 9.5 \approx 570$ V, so $V_\mathrm{peak} \lesssim 1.2\, V_0 + \Delta V \approx 1.75$ kV < 2.2 kV. Snubber discharge at turn-on: $976/10 \approx 98$ A superimposed on the load $\mathrm{d}i/\mathrm{d}t$ — negligible for a disc device. Resistor power:

$$P_R \approx f C_\mathrm{s} V_0^2 = 50 \times 0.5 \times 10^{-6} \times 976^2 \approx 24\,\mathrm{W} \tag{8}$$

per pair; a 50 W pulse-rated wirewound resistor is adequate. Note that after bypass the snubber still sits across the (open) pair in parallel with the closed contactor, so it is unstressed in run; without bypass it dissipates P_R continuously.

8.4 Series strings in MV starters: grading networks

In an MV starter each phase switch is a string of N pairs, and the total blocking voltage must divide acceptably among devices that are not identical. Two networks are used per device (or per pair):

- **Static grading resistor** R_p (tens to hundreds of kΩ) in parallel with each device, sized so that its current is 3–10 times the maximum spread of device leakage currents. It fixes the off-state (DC and 50 Hz) sharing.

- **Dynamic grading = RC snubber** across each device. During commutation, devices recover with different charges; the device that recovers first must absorb the voltage until the others follow. The sharing error is

$$\Delta V \approx \frac{\Delta Q_{\mathrm{rr}}}{C_{\mathrm{s}}}, \tag{9}$$

  where ΔQ_{rr} is the maximum recovered-charge mismatch within the string (from datasheet spread or from lot selection). Equation (9) typically dominates the choice of C_{s} at MV: with $\Delta Q_{\mathrm{rr}} = 2\,\mathrm{mC}$ and an allowed imbalance of $500\,\mathrm{V}$, $C_{\mathrm{s}} \geq 4\,\mu\mathrm{F}$ would be needed — which is why MV designs select/match devices for recovered charge, accept larger per-device margins, or use larger snubbers than the $\mathrm{d}v/\mathrm{d}t$ criterion alone would suggest.

The same string design rules as in HVDC valves apply in miniature: firing delay spread converts directly into transient overvoltage on the last device to fire, which the snubber must buffer; hence the preference for fiber-optic gating with tightly controlled delays at MV.

8.5 Varistors and protection coordination

MOVs across each pair (LV) or each device/string section (MV) clamp line-borne surges. Coordination is straightforward:

$$V_{\mathrm{clamp,max}} < V_{\mathrm{DRM}} \quad \text{and} \quad V_{\mathrm{MOV,cont}} > \hat{V}_{\mathrm{LL}} \times \mathrm{margin}, \tag{10}$$

with the RC snubber handling the fast front (limiting $\mathrm{d}v/\mathrm{d}t$) and the MOV the energy. In addition, overvoltage-triggered firing (break-over diode or active crowbar gating) is often implemented at MV: if the string voltage approaches the design limit, the thyristors are deliberately fired into conduction, which is a safe state for a device feeding a motor.

8.6 Practical remarks

- **Leakage path.** The RC network conducts at line frequency even with gating inhibited: the motor terminals are never galvanically dead through a soft starter. An isolating contactor or disconnector upstream is mandatory for safe maintenance, and this is reflected in the product standards.
- **Capacitor technology.** Film capacitors (dry, self-healing, AC-rated) with low ESL; the loop starter–snubber must be short because stray inductance inside the snubber loop defeats its purpose.
- **Resistors.** Pulse-withstanding types; the peak pulse power V_0^2/R_{s} (tens of kilowatts for microseconds) is the sizing quantity together with the average P_R.
- **No snubber is not an option.** Unlike IGBT converters, where snubberless designs are common, phase-control thyristors in direct line service without RC networks will sooner or later fire on a network transient. The snubber here is a protection element, not an efficiency optimization.

9 Soft Starter versus VFD

The choice at high power is usually between a soft starter and a medium-voltage VFD:

- If the process is genuinely fixed-speed, the soft starter wins on cost (roughly 3–10 times cheaper), footprint, efficiency (after bypass the path is metallic), reliability, and harmonic behavior (transient emission only).

- If the load can exploit variable speed (throttled pumps and fans), the VFD's energy savings normally dominate the economics despite the higher cost.
- The VFD starts the motor at rated flux with up to rated torque at $\leq I_N$ line current; the soft starter fundamentally cannot provide high torque at low current (Eq. (2)). For high-inertia or high-breakaway loads at MV, this is often the deciding technical argument.

A hybrid practice at MV is one VFD used as a *starting drive* for several motors, synchronizing each motor to the line and transferring it; the soft starter is the simpler and far more common realization of the same "start-then-bypass" idea.

10 Conclusion

The thyristor soft starter is a minimal, robust, line-commutated AC voltage controller that solves the starting-transient problem of fixed-speed induction motors from tens of kilowatts (LV catalogue products up to about 1 to 2 MW) to tens of megawatts (MV designs with series-connected thyristor strings). Its power semiconductors operate in a short-time, repeated duty defined by IEC 60947-4-2 utilization categories, and are almost always bypassed in run. Snubbers are an integral protection element in these applications: RC networks across each anti-parallel pair limit the reapplied $\mathrm{d}v/\mathrm{d}t$ at every line commutation, damp recovery-driven ringing against the supply inductance, and — in MV series strings — provide the dynamic voltage grading without which series operation is not feasible. Their sizing is undemanding at LV (fractions of a microfarad, tens of ohms, tens of watts) but becomes a first-order design constraint at MV, where recovered-charge mismatch sets the capacitance and device matching becomes part of the snubber design.

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